

Chapter 15: Rectilinear Motion (Part 1)

TEACHER ANSWER KEY (Detailed)

Grade 10 Physics

Problem 1: Calculation

Step 1: Nature of Motion

The motion is a **Uniform Rectilinear Motion (URM)** because the trajectory is a straight line ("rectilinear path") and the speed is constant.

Step 2: Acceleration

Since the velocity is constant, there is no change in speed. Therefore, the acceleration is strictly zero: $\mathbf{a} = \mathbf{0} \text{ m/s}^2$.

Step 3: Equation of Motion

Assuming the car starts at the origin ($x_0 = 0$), the position-time equation is:

$$\mathbf{x(t)} = \mathbf{v \cdot t}$$

Step 4: Solve the Problem

We need to find the time t required to travel $x = 100 \text{ km}$ at a velocity $v = 80 \text{ km/h}$.

$$t = \frac{x}{v} = \frac{100 \text{ km}}{80 \text{ km/h}} = \mathbf{1.25 \text{ hours}}$$

Problem 2: From Data to Graph

- a) **Graph Plotting:** (See grid on the right).
Points are correctly placed and connected by a straight line.

- b) **Equation of line:**

The mathematical equation is of the form
 $x = v \cdot t + x_0$.

- c) **Original Position and Speed:**

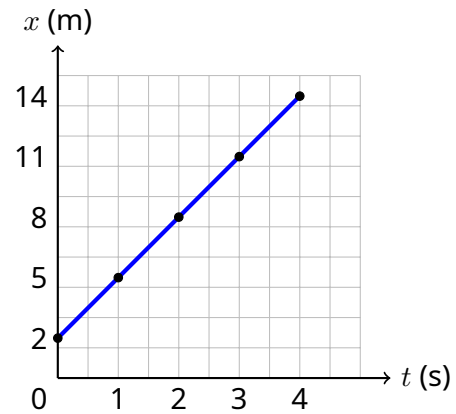
From the table (at $t = 0$), the initial position is $x_0 = 2$ m.

The speed is the slope of the line. Taking two points (0, 2) and (1, 5):

$$v = \frac{x_2 - x_1}{t_2 - t_1} = \frac{5 - 2}{1 - 0} = 3 \text{ m/s}$$

- d) **Nature of motion:**

The $x - t$ graph is a straight oblique line, indicating that the position changes at a constant rate. Therefore, the velocity is constant, making it a **Uniform Rectilinear Motion (URM)**.



Problem 3: From Graph to Characteristics

- a) **Initial position x_0 :**

By reading the vertical intercept (at $t = 0$), the initial position is $x_0 = 1$ m.

- b) **Slope of the graph:**

Choosing the points (0, 1) and (2, 5) from the graph:

$$\text{Slope} = \frac{\Delta x}{\Delta t} = \frac{5 - 1}{2 - 0} = \frac{4}{2} = 2 \text{ m/s}$$

This physical quantity represents the **velocity (v)** of the mobile.

- c) **Equation of motion $x(t)$:**

Substitute v and x_0 into the general formula $x(t) = v \cdot t + x_0$:

$$\mathbf{x(t) = 2t + 1} \quad (\text{SI units})$$

- d) **Nature of the motion:**

The position-time graph is an oblique straight line. This proves the velocity is constant, so the motion is **URM**.

- e) **Position at $t = 1$ s:**

Using the equation of motion: $x(1) = 2(1) + 1 = 3$ m.

(This can also be verified graphically by finding the x -value corresponding to $t = 1$ on the line).

Problem 4: Multi-Phase Motion

a) **Position at the end of Phase 1:**

The mobile starts at $x_0 = 0$ and moves at $v_1 = 4$ m/s for $t_1 = 5$ s.

$$x_1 = v_1 \cdot t_1 + x_0 = 4(5) + 0 = \mathbf{20\text{ m}}$$

b) **Velocity during Phase 2:**

The problem states the mobile "stops". Therefore, its position remains constant at 20 m, and its velocity is $\mathbf{v_2 = 0}$ m/s.

c) **Velocity during Phase 3:**

The mobile returns to the origin ($x_{final} = 0$ m) from its current position ($x_{initial} = 20$ m) in a duration of $\Delta t = 4$ s.

$$v_3 = \frac{x_{final} - x_{initial}}{\Delta t_3} = \frac{0 - 20}{4} = \mathbf{-5\text{ m/s}}$$

(The negative sign indicates the mobile is moving in the opposite direction of the positive x-axis).

d) **Position-Time Graph ($x - t$):**

Note: The total time is Phase 1 (5s) + Phase 2 (3s) + Phase 3 (4s) = 12s. The max position is 20m.

